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Project Documentation

„Project Lab Embedded Systems“

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27

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Project:

**“Development of an Intelligent Capacitive
Humidity Sensor System for Pharmaceutical
Storage“**

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1 Abstract

Sensors that provide reliable monitoring of the environmental condition play an important role in pharmaceutical storage, as any variations in relative humidity will affect the stability of drugs and their expiration time. The present research aims to describe the design, simulate and validate a capacitive humidity sensing circuit for pharmaceutical storage, based on a diode-steered NE555 astable oscillator. It uses changes in the capacitance of CORECI HUMICOR 5000 relative to humidity levels in order to produce an oscillation frequency proportional to it. LTspice simulations of the presented sensor were done across the whole range of its values from 0 to 100% relative humidity and the frequency response was compared to a reference circuit, previously described and experimentally validated with the same sensor and configuration. Software stage of linearization of nonlinear frequency response was performed in MATLAB and resulted in 7.03 %RH maximal error of relative humidity measurement within all of the sensor's range and improved accuracy in 10-70% RH range. Moreover, a script for monitoring was made to show how the obtained humidity information could be used in the monitoring process including trends detection, desiccant lifetime prediction and risk level advisories. This work does not include physical manufacturing of sensors.

2 Member Responsibilities

Mehul Chawla	Linearization in Matlab & Project Documentation
Suhan Roychowdhuri	Algorithm Design & Circuit Construction
Mohammed Abdul Gaffar	Circuit Construction, Report & Final Presentation
Briteen Rajesh Chockar	Final Presentation

3 Functional Description

3.1 Introduction

RH measurement with accuracy is of utmost importance in a range of production and preservation processes, especially those that require controlled environments to maintain the quality of the products being manufactured. In drug storage, changes in humidity may cause a problem in product quality, which consequently spurs continuous efforts towards developing accurate and affordable humidity sensor circuits. The different types of humidity sensors are classified depending on their working principle; however, the commonly used methods of operation are resistive and capacitive. The advantages and disadvantages of each type vary in terms of sensitivity, linearity, response time, and environmental drift. From the recent literature, it is evident how the pros and cons of each type have influenced the design of the circuits.

The construction of capacitive humidity sensors can be performed with the use of polymer dielectrics since there exist some special polymers that demonstrate a significant change of capacitance depending on the humidity of the surrounding air, and thus become applicable to control systems [1]. The usage of such sensors in an oscillator circuit allows one to obtain the

output signal with a predictable change of frequency depending on humidity due to the almost linear reaction of the sensor to humidity variations; this is what the principle of operation of CORECI-type sensors is based upon.

Some research works have focused on alternative circuits and sensing methodologies for measuring the humidity using capacitance. This literature review [2] on floating capacitance measurement methods between 1 pF and 1 nF discusses traditional technologies including lock-in detection and relaxation oscillators, along with modern digital interfacing methodologies like sigma-delta and dual slope converters. All these references show that the method of frequency conversion via oscillation used in the current project is just one of the existing technologies for the conversion of capacitance to signals.

On the sensor-structure interface, [3] studies Kirigami structures obtained from the Japanese art of paper cutting as an attempt to fabricate unique mechanical and functional features through cutting patterns in capacitive sensors. In another direction, [4] examines planar IDTs having polyimide sensing films with different thicknesses and morphologies, finding out that planar IDTs are purely dependent on the ratio of electric field lines passing through the upper layer of the polymer film. Following this trend, [5] expands the technique even further by applying a SAW interdigital electrode in conjunction with a TiO₂ sensing film deposited using the sol-gel process for obtaining an ultrahigh-speed response, with their study revealing no interference from humidity on the SAW signal at 1 kHz, which particular SAW TiO₂ system was not available in the literature before.

Collectively, however, these studies indicate that in recent research there is an increasing trend of focusing on the physical structure of the sensor via different geometries, novel dielectrics or substrate material transducers to improve response time and stability, often alongside advanced electronic signal conditioning circuits. In contrast to such approaches, the solution proposed in the current project takes a simpler approach, whereby the humidity sensor is assumed to be an off-the-shelf capacitor-like device whose properties have already been extensively studied, whereas all the design work is done regarding the read-out circuit, rather than the actual fabrication of the sensor and its characterization. This kind of approach corresponds well to the software and simulation focus of the current project, which does not include any modelling of the sensor. For this reason, the diode steered NE555 astable oscillator circuit used in this project, borrowed from [1], was chosen since it allows one to have a very good experimental characterization of the relationship between the sensor capacitance and the output frequency for the CORECI HUMICOR 5000 sensor.

3.2 Overview

This section provides a succinct and practical description of the system that is meant for those who do not participate in designing it. It describes the design of the circuit, simulation duplication, and the performance of the software associated with the installation of a working sensor.

3.3 Sensor Behavior

This device is built around a capacitive humidity sensor, whose dielectric changes due to the presence of moisture in the air. With an increase in relative humidity of the air, the moisture-absorbing hygroscopic dielectric expands, and there is a corresponding rapid change in its

capacitance value [1]. This effect causes the output frequency of the oscillating circuit, which incorporates this humidity sensor, to drop with increasing humidity, while it increases with decreasing humidity. Due to this reverse and roughly linear relationship between frequency and humidity, humidity can be measured electrically by measuring frequency only.

3.4 Circuit

This circuit is an implementation of a diode-steered NE555 astable oscillator where charging and discharging of the timing capacitor occurs through the NE555 OUTPUT pin instead of the internal DISCHARGE pin. The charge/discharge process is separately controlled by using two resistors-diodes combinations ($R1 = R2 = 18 \text{ k}\Omega$; D1-D4). This design, [1] which is directly based on Chita (2018), was chosen because it gives an output waveform with a duty cycle close to 50%, giving us higher predictability and easier to characterize compared to the usual NE555 astable circuit which provides an asymmetric duty cycle. The sensor (CSENSOR) is connected in series with a capacitor of 30 nF ($C1$) between the TRIG/THRES terminal and the GND. These capacitors make up the timing network whose total capacitance defines the oscillation frequency. Capacitor $C2$ of 10 nF is attached to the CV terminal only to provide decoupling, it has no effect on the oscillation frequency.

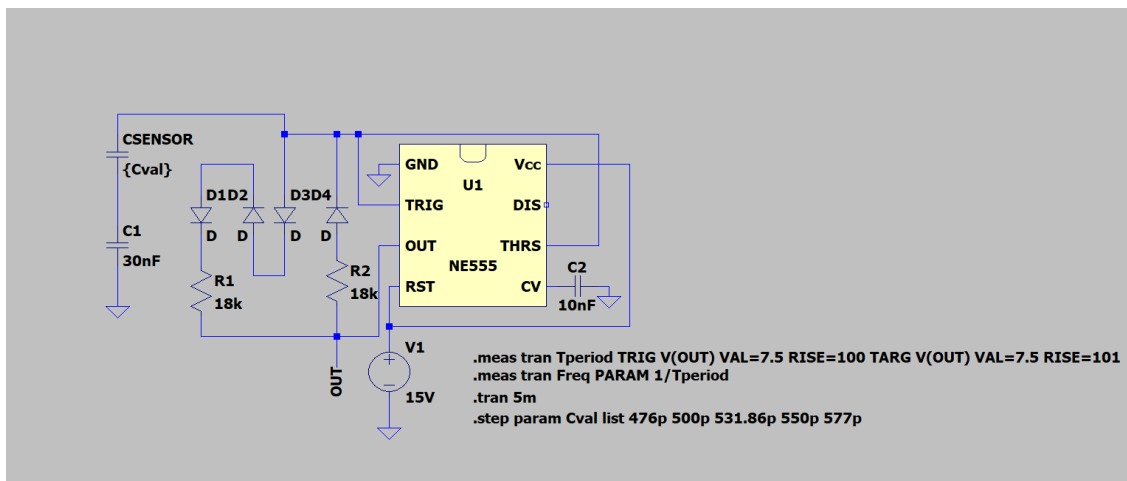


Fig. 1. LTspice schematic of the diode-steered NE555 astable oscillator circuit used for capacitive humidity sensing.

Selection of the CORECI HUMICOR 5000 as the sensing element is justified by the sensor's capacitance variation (from 476 pF to 576.6 pF for 0% RH to 100% RH range) along with its properties like sensitivity of $0.86 \text{ pF}/\%RH$, response time of 2 s , and hysteresis of $0.5\% RH$ which have been documented in [1] using the same diode-steered oscillator circuit used here.

Unlike this, however, the general survey of capacitance measuring methods given in [2], which includes the lock-in amplifier method, the relaxation-oscillator method, and the digital interface using sigma-delta or dual-slope methods, was not used in this project. The mentioned methods usually tend to be more generalized, performing floatation measurements over a larger range of capacitances (1 pF to 1 nF) and therefore require more

complex signal conditioning circuits as compared to an oscillator. Since in the present case, the needed range of capacitance and the needed resolution were already known and fairly small (about 100 pF), this extra complexity was deemed unnecessary.

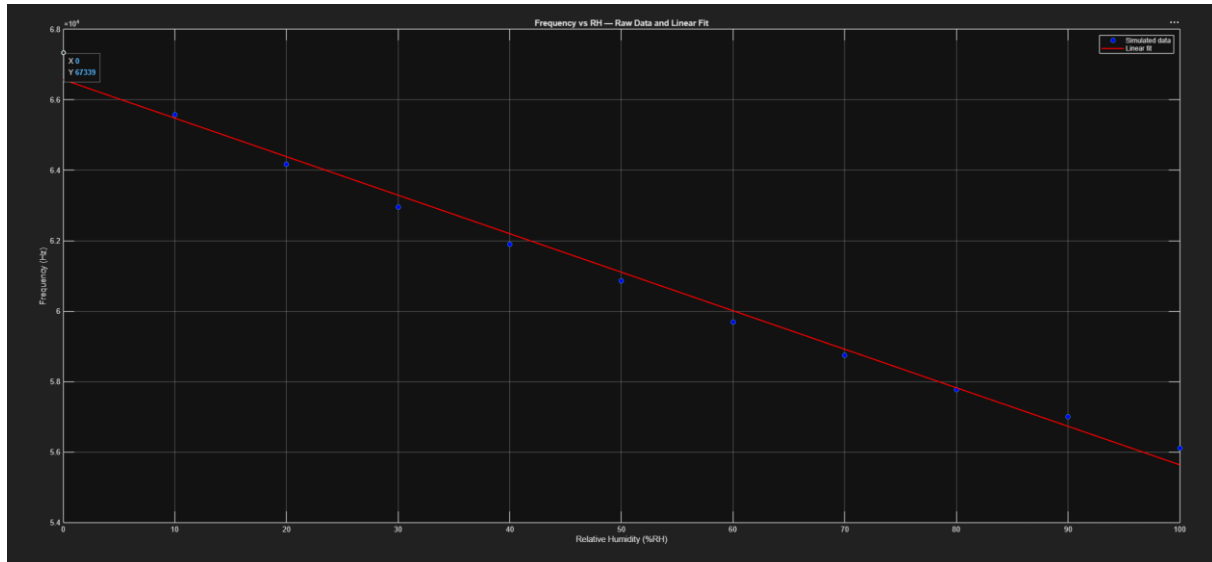


Fig. 2. Simulated oscillator frequency versus relative humidity, with linear regression fit

In particular, the simulated frequency response of the circuit (see Figure Y) was compared to the frequencies obtained in the original paper at each 10% increase in relative humidity (RH), using the same value of 30 nF for C1. In the simulations, there was a consistent difference of 6–7 kHz from the frequencies presented in the original paper throughout all RH values. The reason for such systematic shift could be explained by the differences in SPICE NE555 subcircuit model used in the current simulation compared to the original device or its model in the original study. Regardless of the constant shift, the general shape of the frequency vs. humidity graph resembles the reference graph closely enough, and thus it can be concluded that the key feature of the circuit operation was successfully simulated – monotonic decrease in oscillation frequency as RH increases.

The behavior exhibited agrees well with the performance of capacitive humidity sensors generally observed in literature. According to Islam (2024) [6], for the observed static capacitance-to-humidity relationship, capacitance increases as humidity increases, with the rate being less at low humidity than high humidity and a sensitivity of 1.3 pF/%RH from 5% to 95% RH. The sensitivity is considerably larger than the 0.86 pF/%RH observed in this study with the CORECI sensor which uses TiO₂ thin film with a Surface Acoustic Wave (SAW) based Inter Digital Electrode (IDE) transducer as opposed to the bulk dielectric material used in this study. Independently, [5] studied the Honeywell HCH-1000 polymer film sensor where while observing a linear capacitance response with humidity, he also observed a linear increase in capacitance with temperature. The point of difference is crucial in this regard since in this simulation study, the impact of temperature on the performance of the CORECI sensor was not considered in the framework of the problem statement.

3.5 Software

The exported frequency readings are further analyzed using the supplied MATLAB program, which performs a linear regression analysis on the frequency versus relative humidity data and then inverts this analysis to obtain the relative humidity reading from any frequency reading obtained. The linearization is performed to account for the fact that the output of the oscillator is almost linear in nature although not entirely linear over the 0-100% relative humidity range.

The oscillator signal was linearized using MATLAB through linear regression of eleven data points for frequency and humidity, with subsequent mathematical inversion of the linearization for determining the relative humidity based on the measured frequency. The calibration equation is:

$$Freq = -109.348 \times RH + 66570.362$$

It is demonstrated that the expected negative correlation between oscillator frequency and relative humidity (RH) is indeed valid and conforms to the results obtained in [1].

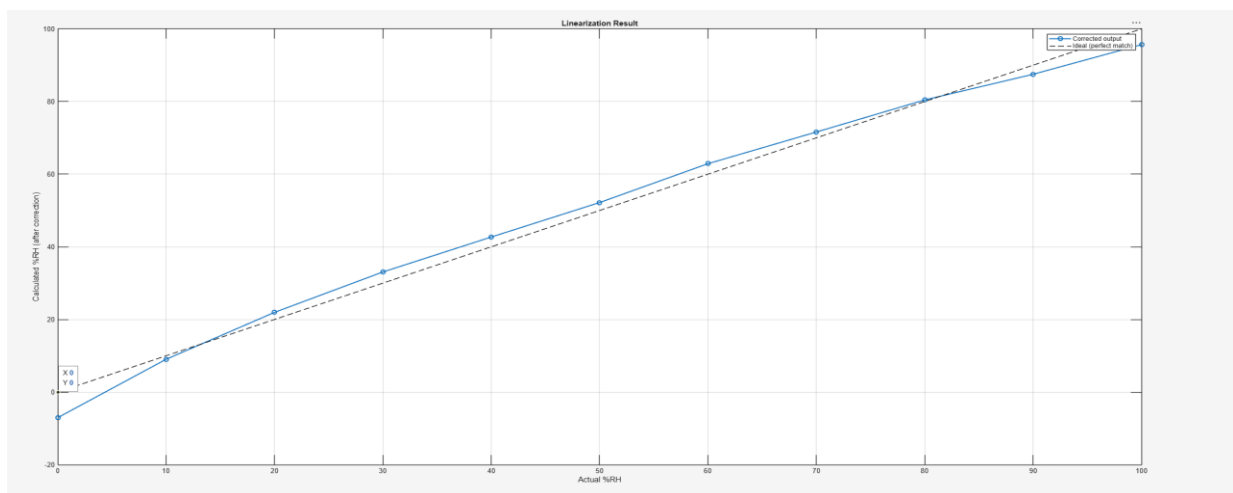


Fig. 3. Linearization result: corrected relative humidity output compared against the ideal (perfect-match) response

In order to determine the degree of validity of the linearization procedure, the corrected output RH value was compared to the actual RH input value for all eleven points used in the simulation. It can be noticed that the error in the correction procedure is non-uniform throughout the measurement interval, as it is maximum at the lower and higher ends of the measurement interval and minimum in the middle of the range.

The maximum linearity error that occurred was 7.03% RH, which was seen at the 0% RH point. This is different from the previous wrongly calculated 1.141%, which was given as a percentage error in the frequency domain instead of %RH. It is important to note that while a certain percentage error may be minimal in the frequency domain, after converting to %RH,

it will not be a percentage error but an error in %RH. The value used for the rest of this discussion of accuracy is 7.03% RH.

Table 1: Linearization error at each simulated relative humidity point.

Actual RH (%)	Linearity Error (%RH)
0	-7.0290
10	-0.9616
20	+1.9498
30	+3.0730
40	+2.6939
50	+2.1885
60	+2.8855
70	+1.6053
80	+0.4355
90	-2.4993
100	-4.3416

This error pattern is consistent with the resolution and accuracy characteristics discussed in the relevant literature. In particular, reference [1] indicates that the linearity error characteristic ($\pm 2.5\%$ RH) is still applicable only to the 10–70% RH range because of the reduced accuracy expected elsewhere; a behavior consistent with the data presented in this paper because the highest measurement errors are observed at 0% and 100% RH, which lie out of the specified range. In turn, reference presents the results obtained with a different (digital, on-chip) [7] humidity sensor design; the estimated measurement accuracy is $+6/-5\%$ RH within the 10–90% RH range taking into account non-linearity, hysteresis and quantization error sources. Thus, the maximum 7.03 %RH error obtained due to linear fit residuals can be considered comparable to the accuracy characteristics of completely different types of humidity sensing devices, which implies the presence of a natural amount of endpoint errors in this type of measurements.

3.6 Simulated Monitoring Behavior

Apart from the pipeline based on LTspice/MATLAB, which is used to verify the circuit, a discrete monitor was developed, illustrating how data conversion from frequency to humidity will work in case of constantly working system, which monitors some kind of pharmaceutical storage. In contrast to the pipeline, which works with real hardware, the script generates possible sets of frequencies and temperatures and processes this data as if it were real: incoming frequency measurements are smoothed using moving average filter to reduce the effect of noise, converted into percentage of relative humidity and compared to the value of previous iteration in order to determine the trend of change (for example, growing, falling or staying constant). The script estimates the remaining lifespan of the desiccant, since it diminishes faster in high humidity conditions, and incorporates all the gathered information about humidity, temperature and the state of the desiccant into a health index and a storage risk category. Based on this analysis, the script outputs the status reports of the system in a form of a brief summary of the humidity and its trends, temperature, risk of damage to stored products, desiccant's lifespan and proper warning message.

Initialization:

- The program starts by importing certain modules, such as time, datetime, collections. deque, and math. At the same time, there is a definition of certain constants.
- Another constant is called WINDOW_SIZE, and it equals 3. It means that there will be a rolling window containing three latest measurements of frequency.
- To be precise, CHECK_INTERVAL has a value of 3. It means that during the execution of the program, the loop will take breaks for three seconds between iterations.
- DESICCANT_LIFE has a value of 45.0, meaning that the initial lifetime of the desiccant pack is 45 percent.
- Another constant is called WINDOW_SIZE, and it equals 3. It means that there will be a rolling window containing three latest measurements of frequency.

Simulation of Data:

- In each iteration of the loop, the step counter increases by one.
- Frequency simulation: The wave is used for simulating frequency, which is then multiplied with time step to generate actual values of the frequency.
- The minimum value for frequency is kept at 8000.0 Hz since frequency sensors have a threshold range of 8 kHz to 13 kHz.
- Temperature simulation: The cosine wave is used for simulating a temperature profile having a center value of 26.0 °C.

Humidity Conversion:

- Frequency value is fed to the buffer.
- The function `smooth_freq` is used to reduce the fluctuations in the frequency values in the buffer, providing a stable value.
- This process is meant to help with reducing any false spikes in the frequency value.
- Later, `calc_humidity` is used to convert the frequency into a relative humidity percent. The outputted relative humidity is ensured to be in the range [0.0, 100.0].
- Firstly, the frequency value is converted to the value of the capacitance.
- Since the raw capacitance values may lie beyond the 0.0-100.0 range, we apply the calibration multiplier to the raw values.
- If the reading from the sensor is small, then the value will be capped at the maximum value of 0 or the negative number.
- On the other hand, if the humidity values are large, the value will be capped at the minimum of higher values or 100.

Trend Analysis:

- The relative humidity is compared to the previous cycles' values.
- It falls under one of the six possible trends.
- which include Rising, Rising Rapidly, Falling Rapidly, Falling, or Stable.
- The relative humidity is then replaced by the new value for the next cycle.

Resource Depletion and Health Scoring:

- The quality of the condition is measured through whether the humidity level is above 60.0% and the temperature level is above 30.0°C, and in such situations, they are regarded as suboptimal.
- The lifetime of the desiccant is used up faster when there is increased humidity, because of the effects of moisture on drying compounds.
- A health measure from 0 to 100.0 is arrived at after deduction of penalties for high humidity and temperature levels.

Risk Assessment and Advisory:

- The metrics are measured against set criteria for categorizing risk levels as follows:
- Critical Danger, Moderate, and Low.
- Depending on the risk level and humidity trends identified, appropriate safety guidelines are issued.

Output and Loop Control:

- The information that we collect is used to make a report containing the current time, humidity along with the trend, temperature, whether the container is open or closed
- the risk involved, the life percent left for the desiccant, and the required guidance.
- Then we use `time.sleep` to halt execution for `CHECK_INTERVAL`, which is 3 seconds.

3.7 Future Scope

While the present work is scoped as a simulation and software validation study, several extensions could move this project closer to a deployable pharmaceutical storage monitoring system.

1. **Modeling of sensor.** In the current model, the CORECI HUMICOR 5000 sensor is modeled as a black box capacitive component using the experimental data of capacitive characteristics in terms of humidity, while a new design could use the modeling of material behavior from first principles. Thus, it would be possible to make the predictions about the behavior of the circuit, using the sensors or dielectrics which were not analyzed before in terms of their behavior. Such approach to modeling of material behavior corresponds to other trends, discussed in recent literature, e.g., [8] states that the future designs of humidity sensors should go beyond single material and empirical calibrations and move toward the integrated design approaches, because the performance of these devices depends on highly nonlinear parameters, such as humidity, temperature and material aging, among others.
2. **Temperature compensation.** In terms of sensor modeling, [9] indicates that temperature dependence and compensation are among the key problems with capacitive humidity sensors, besides the problem of nonlinearity and cost effective manufacture of the sensor. This problem is very relevant with respect to the temperature sensitivity problem mentioned earlier in this report in relation to the Honeywell HCH-1000 sensor [5]. It follows that the logical next step in this project would involve attaching a small temperature sensor along with the humidity sensor in order to make the MATLAB linearization process temperature compensated.
3. **IoT integration and a monitoring interface.** At the moment, the monitor outputs textual results through the console only. The obvious next step will be the connection of the sensor circuit to the network-connected microcontroller for the transmission of humidity, temperature, desiccant life, and risk levels on the dedicated website or mobile app. Considering that the application in question is the storage of medications, there is no need to have the user interface very complicated, but rather straightforward – showing several clear indicators for easy assessment of the storage situation by untrained personnel.
4. **Historical logging and data-driven recommendations.** Other than real-time monitoring, keeping track of every measurement will allow for identification of patterns beyond single instantaneous measurements. Such patterns can lead to two different paths for future research: firstly, linking environmental parameters with

known medicine stability information in order to identify which medicines stored have higher chances of getting damaged due to long-term humidity or temperature deviation; and secondly, comparing how different kinds of desiccants are depleted in various environments in order to find out which kind of desiccants work best in particular environments. The latter will take some time and generation of a large enough dataset before any kind of useful recommendation will be possible to make based on this data. Nevertheless, the logging system itself could be developed as an extension of the current monitoring system script. Such approach corresponds to the future direction of edge computing and AI-enabled sensing that will allow generating more valuable data out of simple sensor readings [8].

5. **Generalized circuit design.** The current configuration uses a 555 timer IC, which is appropriate for implementing and verifying the reference circuit discussed in [1]. In future designs, an oscillator circuit using an operational amplifier may be used, thus making the read-out circuit independent of the timing characteristics of the 555 timer circuit. This will allow using the same measurement circuit for sensors having different capacitance or sensitivity levels without being bound to use the 555 timer because of its internal fixed reference voltages and timing characteristics. Furthermore, the current design will resolve the problem of simple implementation and low cost raised in [9], as the standard op-amp-based circuit can potentially be used for all the different types of sensors.
6. **Self-calibration and drift correction.** Another possibility, inspired by the increasing use of self-calibrating algorithms in current design of system-level sensors [8], is to make the linearization step in the MATLAB code a self-calibration step. In other words, the relation between frequency and humidity can be recalculated at regular intervals based on a few samples at known humidity (for instance, while replacing the desiccant pack), instead of being done only once in design time. Such an improvement will enable addressing the problem of sensor drift or aging, which is currently not handled in the current one-time calibration scheme, and will get the system closer to its intended goal of sustained operation.

4 Description of files

project_documentation.docx	Project Documentation
schedule_presentation.docx	Schedule Presentation Slides
final_presentation.pptx	Final Presentation Slides
additional_files	-
>diagrams	Flowcharts and Tables
>layouts	Circuit Diagram in LTSpice
>source_code	Source Code for any Python

5 References

- [1] M. Chita, „A Circuit for the Relative Humidity Measurement with Capacitive Sensor,“ *International Journal of Innovations in Engineering and Technology (IJJET)*, pp. 24-29, 2018.
- [2] O. Kanoun, A. Yahia und A. Fendri, „Measurement Methods for Capacitances in the Range of 1 pF–1 nF: A review,“ *Measurement*, 2022.
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- [4] J. Boudaden, M. Steinmaßl, H.-E. Endres, A. Drost, I. Eisele, C. Kutter und P. Müller-Buschbaum, „Polyimide-Based Capacitive Humidity Sensor,“ *Sensors*, p. 1516, 2018.
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- [6] D. Mahaseth und T. Islam, „An Ultrafast Capacitive Humidity Sensor With Excellent Static and Dynamic Characteristics,“ *IEEE Sensors Journal*, pp. 8674-8681, 2024.
- [7] B. Wang, M. Law und A. Bermak, „A low-cost capacitive relative humidity sensor for food moisture monitoring application,“ in *4th Asia Symposium on Quality Electronic Design (ASQED)*, Penang, Malaysia, 2012.
- [8] X. Dong, D. Li, A. Chen und D. Zheng, „Humidity Sensing in Extreme Environments: Mechanisms, Materials, Challenges, and Future Directions,“ *Chemosensors* 14, pp. 1-20, 2026.
- [9] M. Sajid, Z. Khattak und K. Rahman, „Progress and future of relative humidity sensors: a review from materials perspective,“ *Bull Mater Sci* 45, 2022.

6 Annex

6.1 Source Code ->

```
import time
from datetime import datetime
from collections import deque
import math

CHECK_INTERVAL = 3
DESICCANT_LIFE = 45.0
WINDOW_SIZE = 3

freq_buffer = deque(maxlen=WINDOW_SIZE)
last_rh = None
sim_step = 0

def get_raw_frequency():
    global sim_step
    sim_step += 1
    wave = math.sin(sim_step * 0.3) * 2000.0
    freq = 11000.0 + wave
    return max(8000.0, freq)

def get_temperature():
    temp_wave = math.cos(sim_step * 0.2) * 8.0
    return round(26.0 + temp_wave, 2)

def calc_humidity(freq):
    capacitance = freq * 0.015
    rh = min(max(capacitance * 0.35, 0.0), 100.0)
    return rh

def check_trend(current_rh):
    global last_rh

    if last_rh is None:
        trend = "Stable"
    elif current_rh > last_rh + 0.5:
        trend = "Rapidly Rising"
    elif current_rh > last_rh + 0.1:
```

```

        trend = "Rising"
    elif current_rh < last_rh - 0.5:
        trend = "Rapidly Falling"
    elif current_rh < last_rh - 0.1:
        trend = "Falling"
    else:
        trend = "Stable"

    last_rh = current_rh
    return trend

def get_risk_level(rh, temp, health):
    if rh > 65.0 or temp > 33.0 or health < 50.0:
        return "CRITICAL DANGER"
    elif rh > 55.0 or temp > 28.0 or health < 75.0:
        return "Moderate"
    return "Low"

def give_advice(risk, trend):
    if risk == "CRITICAL DANGER":
        return "🚨 EMERGENCY: Conditions are bad! Move items and change the desiccant pack!"
    elif risk == "Moderate":
        if "Rising" in trend:
            return "⚠️ CAUTION: Humidity is creeping up. Keep an eye on it."
        return "📌 NOTICE: Getting close to threshold limits."
    return "✅ Looking good. Conditions are stable."

def main():
    global DESICCANT_LIFE
    print("--- Starting Up Environmental Monitor ---")

    try:
        while True:
            timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

            raw_freq = get_raw_frequency()
            temp = get_temperature()

            freq_buffer.append(raw_freq)
            smooth_freq = sum(freq_buffer) / len(freq_buffer)
            rh = calc_humidity(smooth_freq)

```

```

trend = check_trend(rh)
storage_status = "Suboptimal" if (rh > 60.0 or temp > 30.0) else
"Optimal"

usage = (rh * 0.002) if rh > 60.0 else (rh * 0.0005)
DESICCANT_LIFE = max(0.0, DESICCANT_LIFE - usage)

rh_pen = max(0.0, rh - 50.0) * 0.8
temp_pen = max(0.0, temp - 25.0) * 1.0
health_score = max(0.0, 100.0 - rh_pen - temp_pen +
(DESICCANT_LIFE * 0.1))

risk = get_risk_level(rh, temp, health_score)
advice = give_advice(risk, trend)

print(f"\n[{timestamp}]")
print(f" * Humidity:    {rh:.2f} %RH (Trend: {trend})")
print(f" * Temp:         {temp:.2f} °C")
print(f" * Status:       {storage_status}")
print(f" * Risk Level:   🟡 {risk} 🟢")
print(f" * Desiccant:    {DESICCANT_LIFE:.2f} %")
print(f" * Advice:       {advice}")
print("-" * 65)

time.sleep(CHECK_INTERVAL)

except KeyboardInterrupt:
    print("\nStopped by user. Shutting down.")

```


6.2 Datasheet->

CORECI HUMICOR 5000

Table 2: Capacitive Relative Humidity Sensor Data Sheet

Technical Specifications

	H5000	H5100
Response time 90% of scale for a step change from 11% to 75% RH	4 sec	4 sec
Operating range Humidity Temperature Pressure	0–100% RH -30 to +200°C (-22 to +392°F) 0.04–30 bar (0.6–400 psi)	0–100% RH -30 to +100°C (-22 to +212°F) 0.04–30 bar (0.6–400 psi)
Mixing ratio	250g (8.82oz) water/Kg dry air	
Nominal capacity 75% RH @ 23°C (73°F)	500 pF ± 10%	
Sensitivity 11–75% RH @ 23°C (73°F)	0.86 pF / % RH	
Linearity 11–90% RH @ 23°C (73°F)	± 2.5% RH	
Long term stability (12 months) control @ 11% RH	< 1% at 23°C (73°F)	
Maximum air speed (without protection)	< 20m/sec	
Hysteresis	Typical value = 0.5% RH	
D Factor loss tangent @10 KHz 75% RH @ 23°C (73°F)	Typical value = 0.007	
Supply voltage Peak-to-peak	2.5 V AC DC component < 0.2 V	
Operating frequency range	5–300 KHz	
Protection cap Weight	No 0.1g (0.0004oz)	Yes 1g (0.035oz)

Relative Humidity (%RH)	Capacitance (pF)
0	476.00
10	491.80
20	503.10
30	513.10
40	522.62
50	531.86
60	540.95
70	549.95
80	558.88
90	567.77
100	576.62

Table values represent the modeled capacitance-to-humidity relationship reported for the CORECI HUMICOR 5000, used as the calibration reference for the simulation in this project.

Application in This Project

Circuit Integration	Placed in series with a fixed 30 nF capacitor (C1) between the TRIG/THRES node and ground, forming the timing network of a diode-steered NE555 astable oscillator
Output Signal	Oscillator frequency, inversely proportional to relative humidity
Typical Application	Environmental humidity monitoring, climate-controlled storage (e.g. pharmaceutical storage environments)

Source: Compiled from Monica-Anca Chita, "A Circuit for the Relative Humidity Measurement with Capacitive Sensor," *International Journal of Innovations in Engineering and Technology (IJJET)*, Vol. 11, Issue 2, September 2018. DOI: 10.21172/ijet.112.05. No independent manufacturer datasheet for this sensor was available; all specifications above are drawn from this published, peer-reviewed source.